

PRINCE CHAUHAN

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AI/ML Enthusiast | Python & Backend Developer | Building Production-Style AI Systems

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SUMMARY

Computer Science undergraduate with hands-on experience in Python, AI, and backend development. Skilled in building multimodal RAG systems, LLM-powered automation, and REST APIs using LangChain, Flask, Firebase, and Gemini.

EDUCATION

B.Tech in Computer Science and Engineering

2024–2028

Dr. A. P. J. Abdul Kalam Technical University

SKILLS

Languages	Python, JavaScript, C++, SQL
AI/ML & LLM	LangChain, RAG Pipelines, FAISS, Whisper, CLIP, Prompt Engineering, OpenAI API, Gemini AI, Vector DBs
Backend	Flask, FastAPI, REST API Design, Tkinter
CS Fundamentals	Data Structures & Algorithms, OOP
Web Dev	HTML, CSS, Responsive UI, Web Animations
Cloud & Tools	Firebase Firestore, MySQL, Render Deployment, Git & GitHub, VS Code, Postman

PROJECTS

Business Listings Dashboard

Python, FastAPI, SQLAlchemy, MySQL, React, TypeScript

- Architected a full-stack business intelligence dashboard with a FastAPI + SQLAlchemy backend, designing a multi-stage data pipeline (normalization, validation, deduplication) to process and store 3,000+ business listings in MySQL.
- Built a provider abstraction layer (Open/Closed Principle) supporting multiple data sources, with exact-match deduplication via SHA-256 hashing enforced at the database level to guarantee data integrity under concurrent imports.
- Developed a React + TypeScript dashboard with Recharts visualizations (KPI cards, city/category/source distribution charts) consuming versioned REST APIs, backed by Alembic migrations and a Pytest suite covering pipeline and API logic.

Multimodal Video-RAG Event Extraction Chatbot

Python, Gradio, Whisper, CLIP, FAISS, FFmpeg

- Built a multimodal RAG system that lets users search long videos (1-2 hrs) using natural language, combining audio transcripts from Whisper with visual embeddings from CLIP.
- Implemented automated scene detection and clip merging using FFmpeg with a precise three-second buffer, ensuring highly accurate and watchable result segments.
- Engineered a real-time Gradio interface for efficient video querying, incorporating a caching layer that eliminated redundant processing for repeated searches and reduced query times by 75%.

AI Loan Approval System

Python, Flask, Firebase, Gemini, LangChain

- Developed an **AI-driven loan approval** assistant comprising four coordinated agents, streamlining the verification and underwriting process and improving efficiency by 40%.
- Built an admin panel on top of Firebase where staff can view customer profiles, credit history, and full conversation logs - currently tracking 20 customer records end-to-end.
- Generated PDF sanction letters (XHTML2PDF/ReportLab) with Gmail SMTP delivery for applicants the underwriting agent approved - 16 of 20 test customers (80%) - cutting document turnaround from hours to minutes.

ACHIEVEMENTS

- Earned the Silver Certificate from the **Freedom English Academy (FEA)** for completing communication, soft skills, and career-readiness training.
- Served as a **Naval NCC Cadet** under 2 Delhi Naval Unit (2DNU), participating in leadership drills, discipline training, and community service.